

WALNEY ISLAND, United Kingdom

WMO: 032140

Lat: 54.1250N

Lon: 3.2581W

Elev: 14

StdP: 101.15

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.7	-0.6	-5.3	2.4	0.6	-4.2	2.6	1.1	16.3	5.9	14.9	6.2	2.6	10	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	23.0	18.2	21.0	17.0	19.4	16.2	18.7	21.9	17.6	20.1	16.8	18.8	4.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.6	20.0	16.6	11.8	18.7	15.9	11.3	17.9	53.2	22.0	49.7	20.2	47.2	18.9	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.2	11.0	-3.4	26.2	1.2	2.1	-4.3	27.8	-5.1	29.0	-5.8	30.2	-6.7	31.7		
(5)			-4.1	21.1	1.3	2.0	-5.0	22.6	-5.7	23.8	-6.4	25.0	-7.4	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	5.1	5.1	6.4	8.7	11.5	13.9	15.9	15.9	14.1	11.4	8.1	5.9
	DBStd	4.61	2.45	2.42	2.31	2.44	2.55	2.10	2.03	1.78	1.90	2.38	2.67	2.92
	HDD10.0	676	151	137	113	53	11	1	0	0	1	14	66	129
	HDD18.3	2984	409	370	371	289	213	135	82	78	128	216	306	387
	CDD10.0	746	0	0	2	14	57	117	183	184	123	57	9	1
	CDD18.3	13	0	0	0	0	1	2	6	4	1	0	0	0
	CDH23.3	54	0	0	0	0	2	7	29	15	1	0	0	0
	CDH26.7	5	0	0	0	0	0	0	3	2	0	0	0	0
Wind	WSAvg	5.4	6.1	5.7	5.6	5.0	5.0	4.9	4.9	5.0	5.2	5.9	5.9	6.0
Precipitation	PrecAvg	1079	104	76	76	60	65	67	82	91	99	120	116	116
	PrecMax	1404	173	174	155	107	132	202	168	236	200	245	222	265
	PrecMin	841	10	23	30	16	28	21	40	5	13	33	69	35
	PrecStd	151	40	34	29	23	27	41	30	44	42	50	43	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	10.9	14.8	19.1	22.8	24.4	26.6	25.9	21.8	17.5	14.0	11.9
		MCWB	9.6	8.9	10.1	12.8	17.1	18.6	19.4	20.5	17.8	14.8	13.0	10.9
	2%	DB	9.6	9.7	11.9	15.7	20.0	20.9	23.1	21.9	19.5	16.0	13.2	11.1
		MCWB	8.7	8.4	9.1	11.2	15.1	16.7	18.5	18.0	16.3	14.1	12.0	10.3
	5%	DB	9.0	9.0	10.3	13.6	17.3	18.8	20.8	19.9	17.9	15.3	12.4	10.4
		MCWB	8.0	7.9	8.4	10.3	13.4	15.2	17.4	16.8	15.4	13.6	11.2	9.5
	10%	DB	8.5	8.4	9.3	12.1	15.3	17.4	19.1	18.6	17.0	14.6	11.7	9.6
		MCWB	7.5	7.3	7.8	9.6	12.3	14.6	16.4	16.1	14.7	13.1	10.5	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	9.6	11.2	13.6	17.8	19.6	21.3	20.5	18.4	15.6	13.4	11.3
		MCDB	10.3	10.2	13.3	17.7	22.1	23.2	24.4	23.4	20.9	16.8	13.7	11.6
	2%	WB	8.9	8.8	9.6	11.9	15.5	17.0	18.9	18.4	16.9	14.7	12.4	10.6
		MCDB	9.4	9.4	11.2	14.5	19.3	20.3	22.5	21.0	18.7	15.6	12.9	11.0
	5%	WB	8.3	8.2	8.9	10.9	13.9	15.7	17.7	17.4	16.0	14.0	11.5	9.7
		MCDB	8.9	8.8	9.9	13.1	16.5	18.2	20.4	19.3	17.3	15.0	12.2	10.3
	10%	WB	7.7	7.6	8.2	10.0	12.8	14.9	16.7	16.7	15.2	13.4	10.8	8.9
		MCDB	8.3	8.2	9.2	11.8	14.9	16.9	18.8	18.2	16.6	14.4	11.6	9.6
Mean Daily Temperature Range	5% DB	MDBR	3.4	4.0	4.8	5.7	5.9	5.4	5.2	5.0	5.2	4.5	3.9	3.8
		MCDBR	3.1	3.8	5.9	7.7	9.0	8.1	8.3	7.3	6.5	4.6	3.6	3.7
		MCWBR	3.4	3.3	4.1	5.0	5.9	5.2	5.2	4.6	4.4	3.7	3.8	3.9
	5% WB	MCDBR	3.2	3.4	5.0	6.7	7.9	6.9	7.2	6.1	5.5	4.1	3.5	3.7
		MCWBR	3.6	3.3	3.9	4.7	5.6	4.8	5.0	4.5	4.4	3.9	3.9	4.1
Clear-Sky Solar Irradiance	taub	0.330	0.341	0.368	0.387	0.391	0.388	0.390	0.376	0.365	0.350	0.326	0.317	
	taud	2.294	2.369	2.324	2.299	2.338	2.375	2.374	2.425	2.457	2.445	2.368	2.270	
	Ebn at Noon	609	747	805	839	855	861	851	843	805	729	617	539	
	Edh at Noon	60	77	98	115	116	114	112	100	86	70	55	50	
All-Sky Solar Radiation	RadAvg	0.58	1.26	2.39	3.77	4.87	4.98	4.78	3.88	2.75	1.49	0.73	0.44	
	RadStd	0.07	0.14	0.21	0.35	0.37	0.47	0.48	0.27	0.23	0.16	0.08	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page